

GCP Master Data Engineer

6-Month Exhaustive Training & Certification Program

Advanced Architecture, Hands-on Labs, and MLOps

Abstract

This 24-week program is designed to transform you from a practitioner to a Master Data Engineer on Google Cloud. It progressively builds from core storage to advanced Data Mesh governance, integrating daily theory, weekly service-specific mini-projects, architectural use case analyses, and massive monthly Capstone projects. A tracking timetable is included at the end.

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1 Milestone 1: Core Data Infrastructure & Storage (Month 1)

Objective: Master the foundational storage options and transactional databases.

Week 1: Cloud Storage & Transfer Services

- **Monday:** Object storage fundamentals, Storage Classes (Standard, Nearline, Coldline, Archive).
- **Tuesday:** Lifecycle Management, Versioning, and Retention Policies.
- **Wednesday:** Data Migration: Storage Transfer Service vs. Transfer Appliance.
- **Thursday:** **Hands-on Project:** Deploy a GCS bucket via CLI, configure a 30-day transition to Archive, and set up a Storage Transfer job from a public AWS S3 bucket.
- **Friday:** **Use Case & Architecture:** Design a multi-region disaster recovery storage architecture for a media streaming company serving 10PB of video assets.

Week 2: Relational Databases (Cloud SQL & AlloyDB)

- **Monday:** Cloud SQL Architecture, High Availability (HA), and Read Replicas.
- **Tuesday:** Database Migration Service (DMS) and Private IP connections.
- **Wednesday:** AlloyDB for PostgreSQL: Columnar engine and ML integration.
- **Thursday:** **Hands-on Project:** Deploy an HA Cloud SQL Postgres instance, create a read replica, and simulate a regional failover.
- **Friday:** **Use Case & Architecture:** Architect a zero-downtime migration strategy for a monolithic on-premise ERP system to Cloud SQL.

Week 3: Global Relational Scaling (Cloud Spanner)

- **Monday:** Spanner concepts: Nodes, Replicas, TrueTime, and Strong Global Consistency.
- **Tuesday:** Schema Design: Primary keys, Interleaved Tables, and Indexing.
- **Wednesday:** Spanner performance tuning and query execution plans.
- **Thursday:** **Hands-on Project:** Create a multi-region Spanner instance, design an interleaved schema for an e-commerce cart, and load 10,000 synthetic records.
- **Friday:** **Use Case & Architecture:** Design a globally distributed payment processing ledger that requires 99.999% availability and zero data loss across continents.

Week 4: Wide-Column NoSQL (Cloud Bigtable)

- **Monday:** Bigtable architecture: Tablets, Nodes, and App Profiles.
- **Tuesday:** Row Key Design: Salting, hashing, and avoiding hotspotting.
- **Wednesday:** Performance optimization, garbage collection, and Key Visualizer.

- **Thursday: Hands-on Project:** Deploy a Bigtable cluster, write a Python script to ingest IoT temperature sensor data using a reversed-timestamp row key.
- **Friday: Use Case & Architecture:** Design an ingestion and serving layer for a financial trading platform processing 100,000 stock ticks per second.

CAPSTONE PROJECT - Milestone 1: Build an E-Commerce Multi-Tier Data Backend. Deploy Cloud SQL for user accounts, Spanner for global inventory/checkout transactions, Bigtable for clickstream tracking, and Cloud Storage for product images. Map the VPC peering required to connect them securely.

2 Milestone 2: Data Warehousing & Analytics (Month 2)

Objective: Deep dive into BigQuery optimization, Lakehouses, and Data Sharing.

Week 5: BigQuery Architecture & Core SQL

- **Monday:** BigQuery decoupled architecture: Colossus, Jupiter, Dremel.
- **Tuesday:** Pricing models (On-demand vs. Capacity/Editions) and Slots.
- **Wednesday:** Advanced SQL: Window functions, Arrays, and Structs.
- **Thursday: Hands-on Project:** Query the GitHub public dataset, unnest arrays, and calculate a 7-day rolling average of commits per repository using window functions.
- **Friday: Use Case & Architecture:** Estimate the monthly cost of a marketing analytics workload scanning 50TB daily and recommend an Edition tier.

Week 6: BigQuery Optimization

- **Monday:** Partitioning: Ingestion-time vs. Column-based (Time/Integer).
- **Tuesday:** Clustering: Sorting data for optimal filter performance.
- **Wednesday:** Materialized Views, BI Engine, and Query execution plan analysis.
- **Thursday: Hands-on Project:** Create a massive synthetic table. Compare the execution time and bytes billed between a full scan, a partitioned scan, and a clustered scan.
- **Friday: Use Case & Architecture:** Re-architect a poorly performing multi-petabyte log analysis table that is currently causing high query costs for the security team.

Week 7: Advanced Data Types & Federated Queries

- **Monday:** BigQuery GIS (Geospatial) and BigQuery Search.
- **Tuesday:** External Tables and querying data directly in Cloud Storage.
- **Wednesday:** BigQuery Omni (Cross-cloud analytics on AWS/Azure).

- **Thursday: Hands-on Project:** Create an external table pointing to CSVs in GCS. Write a query joining this external data with native BigQuery GIS taxi-dropoff data.
- **Friday: Use Case & Architecture:** Design a multi-cloud data architecture for a retail company that has historical sales in AWS S3 but wants to analyze them in GCP.

Week 8: Data Lakehouse & Analytics Hub

- **Monday:** BigLake: Unifying data lakes and data warehouses with fine-grained security.
- **Tuesday:** Object Tables: Analyzing unstructured data (PDFs/Images).
- **Wednesday:** Analytics Hub: Secure data sharing and Data Clean Rooms.
- **Thursday: Hands-on Project:** Set up a BigLake table with column-level access control. Publish a dataset to the Analytics Hub for a "partner" organization.
- **Friday: Use Case & Architecture:** Architect a Data Clean Room for an ad agency to merge their CRM data with Google Ads data without exposing raw PII.

CAPSTONE PROJECT - Milestone 2: Enterprise Data Lakehouse Build. Ingest 5 years of retail data into GCS. Create BigLake tables over the data, apply row/column level security policies based on region, create Materialized Views for executive dashboards, and set up a BI Engine reservation.

3 Milestone 3: Batch Processing & ETL Orchestration (Month 3)

Objective: Master Hadoop migration, No-code ETL, CDC, and Airflow.

Week 9: Managed Hadoop/Spark (Cloud Dataproc)

- **Monday:** Dataproc Architecture, Ephemeral clusters, and Preemptible VMs.
- **Tuesday:** Migrating from on-prem HDFS to Cloud Storage.
- **Wednesday:** Serverless Dataproc and Spark job tuning.
- **Thursday: Hands-on Project:** Deploy an ephemeral Dataproc cluster, submit a PySpark job that reads from GCS, aggregates sales data, writes back to GCS, and self-deletes.
- **Friday: Use Case & Architecture:** Design a migration strategy for a 100-node on-premise Cloudera cluster, prioritizing cost reduction and decoupling compute from storage.

Week 10: Visual ETL (Cloud Data Fusion)

- **Monday:** CDAP concepts, Data Fusion architecture, and Private IPs.
- **Tuesday:** Wrangler: Interactive data cleaning and transformation.
- **Wednesday:** Plugins, Macros, and Scheduling pipelines.

- **Thursday: Hands-on Project:** Build a no-code Data Fusion pipeline that reads messy customer data from GCS, uses Wrangler to extract domains from emails, and writes to BigQuery.
- **Friday: Use Case & Architecture:** Evaluate Data Fusion vs. Dataflow for a team of business analysts who need to build daily reporting pipelines without writing Java/Python.

Week 11: Change Data Capture (Datastream)

- **Monday:** CDC concepts and the Datastream architecture.
- **Tuesday:** Source configurations (MySQL/PostgreSQL/Oracle).
- **Wednesday:** Destination configurations (Cloud Storage vs. BigQuery native).
- **Thursday: Hands-on Project:** Set up a Cloud SQL PostgreSQL database, configure logical replication, and build a Datastream pipeline to sync changes to BigQuery in near real-time.
- **Friday: Use Case & Architecture:** Architect a zero-downtime database replication pipeline from an on-premise Oracle database to BigQuery for real-time BI reporting.

Week 12: Pipeline Orchestration (Cloud Composer)

- **Monday:** Apache Airflow concepts: DAGs, Operators, Tasks, and XComs.
- **Tuesday:** Cloud Composer 2 architecture (Autoscaling, GKE underlying).
- **Wednesday:** Cross-service orchestration (Triggering Dataproc, Dataflow, BQ).
- **Thursday: Hands-on Project:** Write a Python DAG that: 1. Checks if a file exists in GCS. 2. Triggers a Dataproc job. 3. Loads the output to BigQuery. 4. Sends an email on failure.
- **Friday: Use Case & Architecture:** Design a resilient, self-healing orchestration layer for a daily banking reconciliation process spanning 15 dependent systems.

CAPSTONE PROJECT - Milestone 3: Automated Batch ELT Pipeline. Use Cloud Scheduler to trigger a Composer DAG. The DAG runs a Datastream CDC sync verification, triggers a Serverless Dataproc Spark job to calculate complex ML features, loads them into BigQuery, and creates a Materialized View.

4 Milestone 4: Streaming & Real-Time Analytics (Month 4)

Objective: Conquer unbounded data, message queues, and real-time visualization.

Week 13: Asynchronous Messaging (Pub/Sub)

- **Monday:** Pub/Sub concepts: Topics, Subscriptions (Push/Pull), Acknowledgements.
- **Tuesday:** Dead-Letter Topics, Retry Policies, and Snapshot/Seek.
- **Wednesday:** Pub/Sub Lite vs. Pub/Sub Standard (Cost vs. Operational overhead).
- **Thursday: Hands-on Project:** Create a Pub/Sub topic and pull subscription. Write a Python publisher to stream mock financial transactions. Configure a Dead-Letter Topic for failed messages.
- **Friday: Use Case & Architecture:** Architect a global telemetry ingestion point for 5 million mobile games, ensuring exactly-once processing semantics.

Week 14: Dataflow & Apache Beam (Basics)

- **Monday:** Apache Beam model: PCollections, PTransforms, ParDo, and Runners.
- **Tuesday:** Dataflow architecture: Streaming Engine, Shuffle, Autoscaling.
- **Wednesday:** Dataflow Templates (Classic vs. Flex Templates).
- **Thursday: Hands-on Project:** Write an Apache Beam pipeline in Python that reads texts, tokenizes words, counts frequencies (WordCount), and runs it on the Dataflow runner.
- **Friday: Use Case & Architecture:** Design a pipeline to anonymize incoming healthcare HL7 messages in real-time before storing them in BigQuery.

Week 15: Dataflow Advanced (Windowing & Time)

- **Monday:** Event Time vs. Processing Time and Watermarks.
- **Tuesday:** Windowing strategies: Fixed, Sliding, and Session windows.
- **Wednesday:** Handling Late Data: Allowed Lateness and Triggers.
- **Thursday: Hands-on Project:** Build a streaming Dataflow pipeline reading from Pub/Sub. Group user clicks into 5-minute Session Windows, allowing for 2 minutes of late data.
- **Friday: Use Case & Architecture:** Troubleshoot a scenario where a streaming dashboard is showing inaccurate daily aggregations due to mobile devices going offline and uploading data late.

Week 16: Business Intelligence & Looker

- **Monday:** Looker Architecture: The Semantic Layer and LookML.
- **Tuesday:** Dimensions, Measures, Explores, and Views.
- **Wednesday:** Looker Studio vs. Looker Core (Self-serve vs. Governed BI).
- **Thursday: Hands-on Project:** Connect Looker Studio (or a Looker trial) to your BigQuery retail dataset. Create a real-time dashboard monitoring streaming sales with drill-down filters.
- **Friday: Use Case & Architecture:** Design the BI access layer for an enterprise where 500 analysts need to query centralized data without creating conflicting metrics (e.g., "Gross Revenue" definitions).

CAPSTONE PROJECT - Milestone 4: Real-Time Fraud Detection. Build a Python script simulating credit card swipes sent to Pub/Sub. Create a Dataflow pipeline that uses a Sliding Window to detect if a card is used in two different cities within 10 minutes, flags it, and streams the alert to BigQuery for live Looker visualization.

5 Milestone 5: Machine Learning Operationalization (Month 5)

Objective: Embed AI into data pipelines and manage the MLOps lifecycle.

Week 17: BigQuery ML (BQML)

- **Monday:** BQML capabilities: Regression, Classification, K-Means Clustering.
- **Tuesday:** Time-series forecasting (ARIMA_PLUS) and Anomaly Detection.
- **Wednesday:** Model evaluation (ML.EVALUATE) and hyperparameter tuning.
- **Thursday: Hands-on Project:** Use BQML to train a logistic regression model on Google Analytics sample data to predict user churn. Evaluate the model's ROC curve using SQL.
- **Friday: Use Case & Architecture:** Architect a system to automatically forecast daily inventory needs for 1,000 retail stores using purely SQL-based pipelines.

Week 18: Vertex AI Foundations

- **Monday:** Vertex AI Ecosystem, AutoML vs. Custom Training.
- **Tuesday:** Vertex AI Model Registry and Endpoint deployment (Online vs. Batch).
- **Wednesday:** Explainable AI (Feature attributions).
- **Thursday: Hands-on Project:** Train an AutoML Tabular model on a cleaned BigQuery dataset. Deploy the model to a Vertex AI Endpoint and send a cURL request for an online prediction.
- **Friday: Use Case & Architecture:** Design a workflow for data scientists to train custom TensorFlow models on GPUs while accessing data securely stored in BigQuery.

Week 19: MLOps & Vertex AI Pipelines

- **Monday:** Vertex AI Feature Store: Online vs. Offline serving.
- **Tuesday:** Kubeflow pipelines and Vertex AI Pipelines orchestration.
- **Wednesday:** Continuous Training (CT) triggers and Model Monitoring (Data Drift).
- **Thursday: Hands-on Project:** Build a simple Kubeflow pipeline (Extract → Train → Deploy). Compile it and run it using Vertex AI Pipelines.
- **Friday: Use Case & Architecture:** Architect an MLOps lifecycle that detects data drift in a production recommendation engine and automatically triggers a retraining pipeline via Cloud Functions.

Week 20: Unstructured Data & AI APIs

- **Monday:** Cloud Vision API and Video Intelligence API.
- **Tuesday:** Natural Language API and Translation API.
- **Wednesday:** Integrating AI APIs within Dataflow pipelines.
- **Thursday: Hands-on Project:** Write a Dataflow pipeline that reads customer support text reviews from Pub/Sub, calls the Natural Language API to get sentiment scores, and saves to BigQuery.
- **Friday: Use Case & Architecture:** Design a scalable pipeline to process terabytes of uploaded user images, extract text (OCR), tag objects, and make the metadata searchable.

CAPSTONE PROJECT - Milestone 5: End-to-End MLOps Pipeline. Use Composer to orchestrate a pipeline that extracts features from BigQuery, pushes them to Vertex Feature Store, trains a custom model using Vertex Training, registers the model, and sets up Vertex Model Monitoring for serving skew.

6 Milestone 6: Security, Governance & Exam Prep (Month 6)

Objective: Secure the perimeter, implement Data Mesh, and certify.

Week 21: Security, IAM & Networking

- **Monday:** VPC Service Controls (Perimeters) and Private Service Connect.
- **Tuesday:** IAM Deep Dive: Custom Roles, Service Accounts, and Workload Identity.
- **Wednesday:** Cloud KMS (CMEK vs. CSEK) in BigQuery and GCS.
- **Thursday: Hands-on Project:** Create a VPC Service Perimeter around a BigQuery dataset. Attempt to query it from an external IP to verify the block. Configure an access level to allow your IP.
- **Friday: Use Case & Architecture:** Design a highly secure data perimeter for a healthcare provider ensuring no data exfiltration can occur even if IAM credentials are compromised.

Week 22: Data Governance & Data Mesh (Dataplex)

- **Monday:** Dataplex architecture: Lakes, Zones (Raw/Curated), and Assets.
- **Tuesday:** Data Catalog: Tag templates, Search, and Data Lineage.
- **Wednesday:** Auto Data Quality (AutoDQ) and Data Profiling.
- **Thursday: Hands-on Project:** Set up a Dataplex Lake. Map a GCS bucket and BigQuery dataset as assets. Create a tag template and apply policy tags to restrict access to a 'Salary' column.
- **Friday: Use Case & Architecture:** Architect a Data Mesh for a global conglomerate with 5 independent domains, ensuring centralized governance but decentralized data ownership.

Week 23: Compliance & Data Loss Prevention (DLP)

- **Monday:** Cloud DLP concepts: InfoTypes, Inspection, and De-identification.
- **Tuesday:** Masking, Tokenization, and Crypto-based hashing.
- **Wednesday:** Automated DLP scanning in BigQuery and GCS.
- **Thursday: Hands-on Project:** Write a Python script using the DLP API to scan a block of text, identify emails and credit card numbers, and replace them with asterisks (masking).
- **Friday: Use Case & Architecture:** Design a pipeline that ingests third-party customer data, automatically identifies PII, tokenizes it using a KMS key, and stores the safe version in the data warehouse.

Week 24: Final Certification Preparation

- **Monday:** Case Study Review 1: EHR Healthcare (focus on compliance and migration).
- **Tuesday:** Case Study Review 2: Helium Sports (focus on real-time analytics).
- **Wednesday:** Full-length Mock Exam 1 (Identify weak domains).
- **Thursday:** Targeted review of weak domains based on mock exam results.
- **Friday:** Full-length Mock Exam 2.

CAPSTONE PROJECT - Milestone 6: Secure, Governed Data Mesh. Build a multi-project architecture. Project A ingests streaming PII data and uses DLP to tokenize it. Project B (Analytics) hosts BigLake tables. Project C (Governance) uses Dataplex to enforce row-level security and VPC Service Controls to lock down all API access.

7 6-Month Progress Tracking Timetable

Week	Primary Focus	Key Project / Capstone	Status
1	GCS & Transfers	GCS Lifecycle & Transfer Job	☐
2	Cloud SQL & AlloyDB	HA PostgreSQL Failover Simulation	☐
3	Cloud Spanner	Global Interleaved Schema	☐
4	Cloud Bigtable	Cap 1: E-Commerce Multi-Tier Data	☐
5	BigQuery Architecture	Advanced SQL Window Functions	☐
6	BQ Optimization	Partitioning & Clustering Benchmarks	☐
7	BQ Advanced	GIS & External Tables	☐
8	Lakehouse & Analytics Hub	Cap 2: Enterprise Data Lakehouse	☐
9	Cloud Dataproc	Spark Job on Ephemeral Cluster	☐
10	Cloud Data Fusion	No-code Wrangler ETL Pipeline	☐
11	Datastream (CDC)	PostgreSQL to BQ Logical Sync	☐
12	Cloud Composer	Cap 3: Automated Batch ELT	☐
13	Cloud Pub/Sub	DLQ and Push/Pull Scripting	☐
14	Dataflow Basics	Beam WordCount on Cloud Runner	☐
15	Dataflow Advanced	Session Windows & Late Data logic	☐
16	Looker & BI	Cap 4: Real-Time Fraud Telemetry	☐
17	BigQuery ML	Churn Prediction & Evaluation	☐
18	Vertex AI Foundations	AutoML Deployment & cURL Predict	☐
19	MLOps & Pipelines	Kubeflow Pipeline Orchestration	☐
20	AI APIs	Cap 5: End-to-End MLOps Pipeline	☐
21	IAM & VPC SC	Service Perimeters & Access Levels	☐
22	Dataplex & Data Mesh	AutoDQ and Policy Tags	☐
23	Cloud DLP	Python Tokenization/Masking Script	☐
24	Exam Prep	Cap 6: Secure Data Mesh Design	☐

"Consistency is the architecture of mastery."

Best of luck on your 6-month journey to becoming a Master Google Cloud Data Engineer!